2022

High School Mathematical Contest in Modeling (HiMCM) & Middle Mathematical Contest in Modeling (MidMCM)

Summary Sheet

Team Control Number: 13296

Problem Chosen: C

This paper explores the optimal schedule for a group of friends at an amusement park, and thoroughly analyzes the relevant factors. The problem required that several assumptions be made before listing the group member's preferences in order to develop an effective priority list. After listing our assumptions and describing an ideal day for each of the group members, we created the schedule. In order to do this we developed a model in which we only considered events that aligned with our parameters, which included various priorities for the group, short wait times, and proximity to the previous event. We used the website, listed in the sources (1), to calculate the distance between different rides, and used $t=d/r$ to find the walk time, the wait time for the event (if applicable), and the amount of time the actual event took. This model led us to a schedule for the given group and can also be used to create a schedule if the variables considered were different.

Next, we wrote a letter to the group of friends describing the schedule and considered our models various strengths and weaknesses. Then, we described our own ideal schedule and how it would differ from the group of friends given.

The schedule we provided is efficient and will be liked by the group. We arranged some of the most popular rides at the beginning of the day where the waiting times wouldn't be a problem. The group splits up so everyone can attend their ideal activities. Ming and Freya visit the arcade, while Karine and Ishamael visit some of the tougher roller coasters. After a morning occupied by rides, they head to the Circus Show. They attend the Triangle Restaurant next where waiting times are shorter. After, the group goes to the Magic Show. This is followed by getting ice cream more rides. Soon it is time to visit the Gift Shop where the friends can purchase gifts for their trip. The day would be concluded with dinner and a view of beautiful fireworks.

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1 Introduction

1.2 Restatement of the Problem

The problem at hand is to optimize a schedule for four friends at a fictional amusement park known as Polygon Paradise which offers a variety of activities. Four friends: Ming, Ishmael, Karine, Freya, all have their own individual preferences for which activities they like and dislike. In order to optimize a schedule one must create a model that effectively and efficiently considers and deals with the many variables involved. These factors include the ride time, wait time, walk time, distance between rides, bathroom breaks, food, preferences, and dislikes. The effective model will take into account these factors and create a system to produce an efficient schedule. The model should then be able to produce an efficient schedule with different parameters then the given problem. It is important to note that the problem of efficient scheduling is an NP problem, meaning that it may not be possible to produce an algorithm for the optimal schedule, so a correct solution can be a nonoptimal but relatively close to an optimal solution.

2 Assumptions

1. **Walking Speed: The walking speed for the group is constant and is 1.3 m/s.**

   Justification: This is the average walking speed given by the NCBI. (2) and this assumption of constant pace will help our task of creating a more straightforward and accurate model.

2. **Bathroom Usage: The group will have to go to the bathroom 3-4 times during the day trip.**

   Justification: According to our sources (3) , the amount of bathroom trips in 24 hours is around 4-10, the average of which is 7. Since the group is only in the park for 12 hours, we can say the average trip is around 4 times. (rounded from 3.5)

3. **Breakfast: The group ate breakfast before arriving at Polygon Paradise Park.**

   Justification: According to Northwestern Medicine, people should aim to eat breakfast within an hour of waking up. A survey conducted by Edison Research shows that most people wake up before 7:30, meaning the group of friends will most likely eat breakfast before 8:30. Our job is to provide a schedule for the group starting from 9:00 AM, so breakfast will not be considered as a factor for our model.

4. **Arrival time: The group arrives at the park before or at 9:00 AM**

   Justification: We are tasked with providing them a schedule starting at 9:00 AM and ending at 9:00 PM. This means that they are expected to arrive at 9:00 AM, and we are **only** responsible for providing the schedule between that 12 hour window.

5. **The ideal lunch time is from 11 AM to 2 PM (4-5 hours after breakfast).**

Justification: The ideal lunch time according to multiple sources is 4-5 hours after breakfast, and 11 AM to 2 PM are the corresponding times assuming a breakfast from 7-9AM. (5)

6. **All measurements of time in minutes will be rounded UP to the nearest minute.**

   Justification: Traveling in an amusement park will never be perfect, and it is completely okay to omit half a minute for traffic and/or getting on rides.

7. **The group would prefer to eat together.**

   Justification: Eating is an inherently social activity and most people prefer to eat with others, as evidenced by a report from the University of Oxford. (4)

8. **Between given hours, wait times change at a constant rate.**

   Justification: It is unlikely that wait times simply jump in time from one hour to the next, so it is most practical logical to assume that wait times decrease at a roughly constant rate.

9. **Each member's preferences matter equally and are weighted in a schedule as such.**

   Justification: We are tasked with creating the most efficient, but also most enjoyable trip for the team. If we disregard the member's preferences, they would have a less enjoyable time at the park, and we are tasked to create a schedule that guarantees that everyone has a good time.

10. **The group would prefer to ride rides at times when the wait times are in the bottom quartile.**

    Justification: As evidenced by Karine's preference, humans are fundamentally impatient. Furthermore, minimizing waiting time allows the group's day to be

maximally efficient. They can do more fun activities and rides instead of waiting in long lines, which is explained in an article cited (6).

11. **Each member of the group will want to try other activities than just their preferred activity.**

    Justification: The group members are not confined to their stated preferences and a complete amusement park is designed to contain variety.

12. **The arcade is open from 9 A.M. to 9 P.M. and has no waiting time.**

    Justification: The information provided doesn't say anything about when the arcade will be open, so we will assume the arcade is open from 9 A.M. - 9 P.M. In an arcade, there are waiting times for specific games, but no information is given about games or waiting times. We won't consider this as part of our model to make it less complex.

13. **It will take roughly 10 minutes for a bathroom trip**

    Justification : One of the articles cited (7), states that it takes around 20 seconds to actually go to the bathroom. However, considering washing hands, flushing toilets, and waiting in lines, we will say it takes around 10 minutes for a bathroom break. Giving lots of time ensures that if the line is longer, it won't affect the whole schedule.

3 Considerations

3.1 Attractions

In Polygon Park everyone can find a perfect activity. This amusement park has every ride you could ever possibly think up of. A great time is guaranteed with all our rides: roller coasters, water rides, and more! A train ride, carousel, and multiple roller coasters are included. This park will also provide non-ride entertainment such as magic shows, an arcade, circus acrobatics, and a fireworks show! The park also has an ice cream stand, a hotdog stand, and a sit-down restaurant. Polygon Park is equipped with every attraction possible for the perfect day at an amusement park!

3.2 Factors to Consider

The preferences of the group members is the most important consideration. First, we must consider that Ishmael wants to see the magic show, which is only playing during two different times. Similarly, Karine wants to see the circus acrobatics show, which shows three times. Karine especially dislikes waiting in long lines, so choosing a time for the show where wait times for other rides are longer would be optimal. Karine also dislikes hotdogs, so if the group eats lunch or dinner together they must eat at the triangle restaurant. Our next consideration is that Ming gets motion sick after going on fast rides, especially after eating. Thus, if Ming must go on a ride it should not be directly after a meal. Finally, the group can't split up for more than 4 hours, so the activities that are done together should be as enjoyable to everyone, and when the priorities for different people clash the group should split up into 2 groups of 2 people. It is important to note that as stated in our assumptions, the schedule must consider and weigh all preferences equally.

3.3 The Ideal Day

Ming:

The problem states that Ming strongly dislikes roller coasters and becomes motion sick easily. However, visiting Ming loves playing games and would probably

like to visit the arcade. Based on this information, it would be ideal for Ming to spend a good amount of time in the park's arcade, perhaps most of his entire day. If he were to go on any rides at all, they would probably be slow-paced and basic. Ming seems okay with eating anything, so it would be okay for him to eat at any of the restaurants.

Karine:

The problem states that Karine wants to attend the circus acrobatics show. On top of that, Karine dislikes waiting in long lines and eating hot dogs. From what we are told from the problem, we know she would prefer to spend an hour watching the circus acrobatics show. Her day would consist of going on rides during which they have shorter waiting times, eating food at the triangle restaurant and ice cream stand, and watching some of the shows available.

Ishmael:

In the problem it is stated that Ishmael loves roller coasters and wants to ride as many of them as possible during the day. Ishmael also likes performing card tricks and is interested in seeing the magic show. Based on this description, one can gather that Ishmael would like to ride on the heptagon coaster, the rhombus ride, and the decagon coaster, all preferably at least once. Ishmael also would also like to try the different food options for lunch, dinner, and an occasional snack and attend other rides and shows not previously mentioned for a varied experience. Lastly, Ishmael would also spend an hour of his day watching the magic show.

Freya:

From the preferences we are given, Freya would like to go to water rides and the souvenir shop. In the assumptions we assumed that each person would like to try other activities as well, so Freya's ideal day would not only include multiple trips to the water rides but other rides and shows as well. She would also like to visit the food vendors to try out the different foods available. Near the end of the visit to Polygon Park, she would buy a souvenir to remember her amazing day at Polygon Park.

4 Model Description

Notation	Definition
t	Amount of Time task takes
s	Walking Speed
y	Wait Time
d	Distance
x	Minutes passed since the current hour
m	Slope
b	Y-Intercept

Throughout our model, the above variables and the groups preferences were primarily considered in our model for determining the ideal schedule for the group. In determining the sequence of events distance was the first parameter considered. Thus, only activities close to the one previously done were considered. The other parameter we considered was the activity's priority, determined by the groups preferences and dislikes. The final parameter we considered was the wait time. Events with excessive wait times at a given time were thus avoided.

4.1 Essential Activities

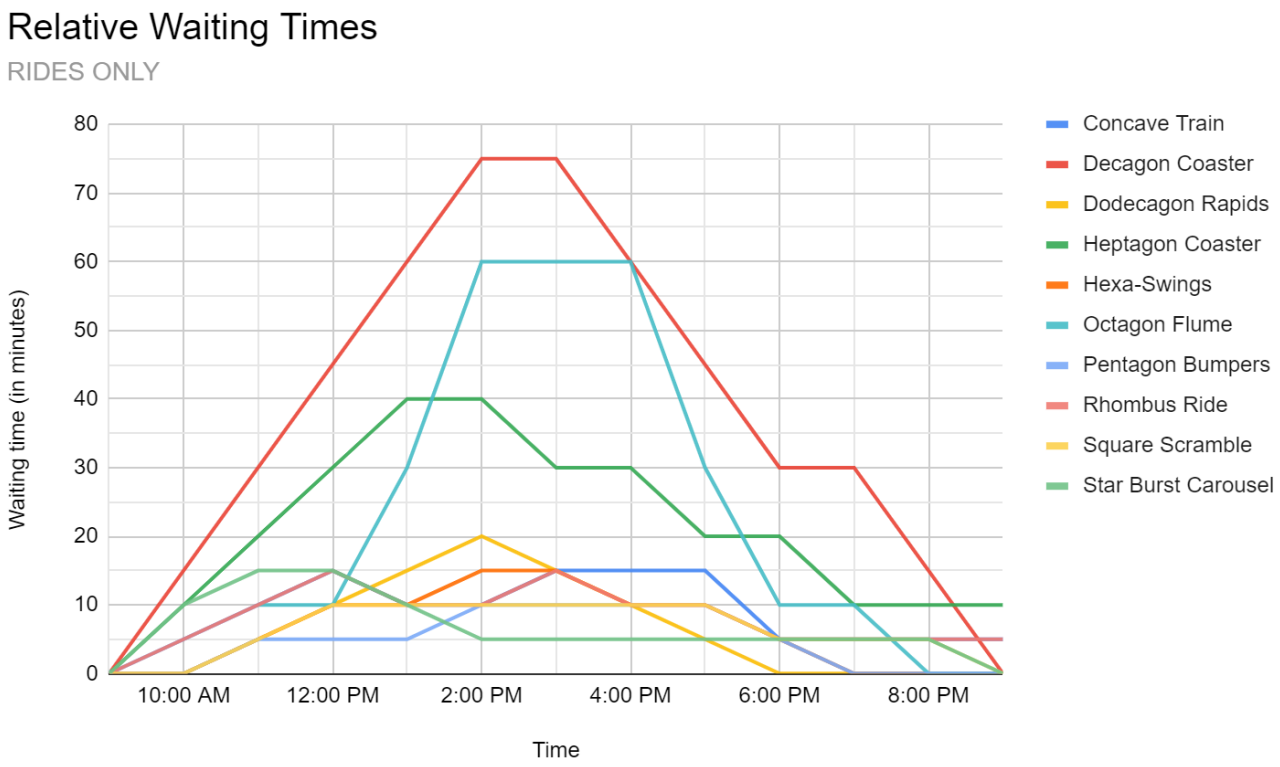
Ishmael	**Ming**	**Karine**	**Freya**
Magic Show	Game Arcade	Circus Acrobatics	Souvenir Shop
Decagon Coaster			Dodecagon Rapids
Heptagon Coaster			Octagon Flumes
Rhombus Ride			

This also includes bathroom breaks and stops at the food options.

4.2 Times to Split Up

For the group to have an optimally efficient day, Ming, Ishmael, Karine, and Freya will need to all achieve what they want to do at the park, along with trying some new activities. This means that the group should split up to dissolve disagreements in preferences. Take example Ming and Ishmael: Ishmael loves roller coasters and would prefer to ride as many as possible, while Ming is easily motion sick and would not be able to handle speedy roller coasters. In this situation, we should take advantage of the time where the group can split up, so that both Ming and Ishmael could achieve their ideal activities throughout the day. However, it is ideal for the group to stay together for the majority of their trip. Finally, it would be logical to conclude that the group should meet at a location that is feasible to arrive at, along with having an activity that is preferable to everybody.

4.3 The Schedule



[1]

First, one has to consider what the first event is. Of the rides mentioned, the ride to prioritize is the Decagon Coaster because it is the ride with the longest wait times, which is shown in the graph above. To avoid longer wait times later, it is optimal to ride it first. While we know Ming gets motion sick on rides, it is not optimal for the group to split up for only one activity. So the group will go together, and Ming could sit this ride out.

If they rush to the Decagon Coaster, they can be among the first in line and thus avoid the long wait time. This will take the group approximately three minutes. We calculated this by using the formula for speed.

$$s=d/t$$

Or

[1] This chart was made in Google Sheets with the information given in the problem.

$$t=d/s$$

Since we know the distance and speed all we can plug it into the equation, and convert our answers into minutes.

$$t = 222.25\text{m}/1.3\text{s}$$

$$t \approx 171s \approx 3min$$

Then, we have to find the wait time at 9:03. From assumption one, we are assuming the wait time increases and decreases linearly. The initial wait time is 0 minutes, and the final time at the end of the hour is 15 minutes. Then will find the equation of this segment that represents the wait time of the hour. If we use the slope intercept form :

$$y=mx+b$$

x represents the time in minutes, for example, if the time is 10:05, x would be 5 since the hour can be omitted from the calculation of the slope. We can omit it because we calculate the slope of the hour, so the hour would stay the same. y represents the wait time.

To find the slope, we can use :

$$m=\frac{y_2-y_1}{x_2-x_1}$$

We can plug the numbers in, m = (0-15)/(0-60) =-15/-60 = ¼, which is the slope. The y-intercept is 0, so the equation will be: $y=(¼)x+0$. If we plug the time in, 3 minutes, so y=¾ which is .75 minutes, and we will round to 1 minute. Thus they will start at 9:04, and with a ride time of 3 min they will be finished at approximately 9:07. This method is how we calculated the wait times of the other rides at a specific time.

The second activity is the Concave Train, since the train is very close to the Decagon Coaster and no one dislikes the ride. The walk time will be around 60 seconds

and the wait time at 9:08 is around 1 min. Thus, the group would start the train at 9:10 and would end at 9:25.

The third activity will be the Hexa-Swings. It is the closest to the Concave Train, and no one in the group specifically dislikes it. The ride takes approximately 1 minute to walk to and has an approximate wait time of 5 minutes around 9:30 AM, so the group starts the activity at 9:31 AM, and ends the activity around 9:33 AM due to the ride's 2 minute duration.

Next, the group will visit the Square Scramble, as it is relatively close to the Hexa-Swings. The walk takes 3 minutes, and the wait time is still 0 minutes. The ride lasts 2.5 minutes. The activity lasts a total of around 6 minutes.

The next activity they will do is the Octagon Flume. This is listed as one of Freya's priorities because it is a water ride. Walking there takes roughly one minute, and the waiting time at 9:40 is roughly 7 minutes. The ride is 5 minutes, and in total the activity lasts 13 minutes.

Starting here on out, the group will part paths into two smaller groups. There are many reasons behind this decision.

The main reason that caused us to make this choice is that Ming dislikes fast moving rides, so having them travel as a big group for the entire day may either prevent Ishmael from riding as many roller coasters as possible or cause Ming to get motion sick or be unhappy.

To best suit all the group members and maximize fun, the group will split up. We can split the group into Karine and Ishmael, Ming and Freya. Karine and Ishmael can go on roller coasters, while Ming and Freya can spend more time at the arcade and do some basic rides. Freya is flexible and doesn't mind most rides, so she would get along well with Ming at the arcade and some easier rides. Karine doesn't like waiting and Ishmael enjoys more intense activities, meaning they would go well together with a vigorous afternoon. By doing this, everybody should be able to complete their preferred activities.

Ishmael and Karine head to the Heptagon Coaster. The walk is approximately 4 minutes, and because it is just before 10:00 AM, the waiting time is just under 10 minutes. The ride lasts 2.5 minutes. In total, they start walking towards this activity at 9:52 and end at 10:09, lasting 17 minutes.

After this, the two will go on the Rhombus ride, since it is relatively close and provides extreme thrill. The walk takes 2 minutes, and the wait time is 6.25 minutes. The ride takes 4 minutes, so in total this activity takes another 13 minutes.

Because Ishmael wants to ride as many thrilling rides as possible, we will take advantage of the short waiting times in Rhombus Ride during this time period and have Ishmael and Karine ride it again. They are already at the Rhombus Ride, so without traveling much they can stand back in line. At 10:22, the wait time will be approximately 7 minutes. The ride lasts 4 minutes, so they will be done at 10:33.

Now, Ishmael and Karine will walk to the bathrooms, where they will meet with Ming and Freya for a bathroom break.

While Ishmael and Karine go on these rides, Ming and Freya will hang out at the game centers and play games, hopefully winning prizes. This will last from 9:52 AM to 10:34 AM. The walk from Octagon Flume, their previous location, to the game center, will last roughly two minutes. They can spend 40 minutes there before starting their short walk, lasting under a minute, to meet the other half of their group for a 10 minute bathroom break.

The bathroom break will go from 10:35 - 10:45.

While the Octagon Flumes are close by and a top priority for Freya, it is further away than the Starburst Carousel and has a slightly longer wait. On top of that, the group has already taken the ride once. We don't want to waste more than 10 minutes waiting for any ride despite its priority, since it would waste time and frustrate Karine. Instead, the group walks the short distance from the game center to the Starburst Carousel, taking roughly 1 minute to get there. The wait time is 8.75 minutes since it is a quarter before 11, and the ride lasts 3 minutes. The activity takes roughly 13 minutes in total.

At this time, the most suitable ride for the group would be the Pentagon Bumpers. It would take approximately 4 minutes to walk there, so the group would arrive around 11:02 AM. From 11:00 AM to 12:00 AM the waiting time remains at 5 minutes. Furthermore, this meets everybody's priorities as it wouldn't make Ming motion sick, it wouldn't make Karine wait too long, and it would give Freya quality time with her friends. The ride has a duration of 5 minutes, meaning that the group would start the ride at 11:07 AM and end it at 11:12 AM. The activity would take a total of 14 minutes.

Afterwards, the group will walk roughly 50 meters to the Dodecagon Rapids. This is one of Freya's priority rides, and the waiting time is short. Not only that, it is also very close by. It takes about a minute to walk over, and at 11:13 AM, the waiting time is roughly 6.25 minutes. The ride itself lasts about 10 minutes. They would start walking towards the ride at 11:12 AM and finish the ride at 11:30 AM. This means the elapsed time of this activity is 18 minutes.

Subsequently, after the lengthy water ride, it may be nice to give the Pentagon Bumpers another go. The walk between the two places takes one minute, and the wait time at 11:30 AM is 5 minutes. The ride itself lasts for 5 minutes. In total, the activity lasts around 11 minutes, from 11:30 AM to 11:41 AM.

Now, whilst anticipating the Circus Acrobatics show that Karine wants to attend, starting in 18 minutes, we have come to the conclusion that the best use of this time would be to play the group's favorite game in the arcade again. Waiting times for rides start to peak and there are no available shows at the time. They had a recent bathroom break and ice cream would not be ideal too soon before lunch. Walking there takes just under 3 minutes, and 13 minutes can be spent at the game center before they have to start walking to the Trapezoid Show. It takes just over one minute to get there, arriving just in time for the Circus Acrobatics Show. In total, the time elapsed in this section of their schedule is approximately 15 minutes.

Upon reaching the Trapezoid Show, the group will sit down and watch the show, which starts at 12:00 PM and ends at 1:00 PM.

Now, after several activities and a show, the group will certainly start getting hungry. Wait times for the Triangle restaurant are going down, and 1:00 PM is definitely a valid time for lunch.

The group will stay together for both lunch and dinner, because from one of our previous assumptions stated people like to eat with each other and have more fun doing so. They will head to the Triangle restaurant because Karine will not eat hot dogs. The walk takes 4 minutes. Once they get to their destination, they must wait 20 minutes before getting their food. This brings them to the time 1:24 PM. The group decides to spend roughly 30 minutes at the restaurant, which we were told was enough to enjoy a meal. They would finish around 1:54 PM.

At this time it would be perfect to catch the next round of the magic show at the Trapezoid Show. This is also quite convenient because the waiting times for rides rise to a peak around this time, so it would be more efficient to watch the magic show

instead of waiting in a long line. Ishmael also has this as a priority, so this would also be a highlight for him. It would take the group approximately 4 minutes to walk to the Trapezoid Show from the Triangle Restaurant, so they would get there just before 2:00 PM. The show ends at 3:00 PM, so this activity takes up just over an hour.

Now the time is 3:00 PM. Ride times are still at quite a peak, and the group has completed much of their to-do list. The best activity possible for the group at this point would be the arcade. Ishmael and Karine have yet to play at the game center, and Ming and Freya only spent 40 minutes there. The group will spend another 45 minutes at the arcade, that way all group members can give the amusement park games a shot. Walking there takes just under 2 minutes, so they will arrive by 3:02 PM. After spending 45 minutes there, they will depart the game center at 3:47 PM.

After the arcade, it will be nearly 2 hours after lunch. This would be a good time to grab a snack. The group will head to the ice cream cart to get some ice cream. They will spend around 2 minutes walking there, and the waiting time for the ice cream cart at 3:49 PM is roughly 11 minutes. Upon getting their ice cream at 4:00 PM on the dot, they can eat it while heading to their next activity.

The next ride on the list is the Starburst Carousel. It is relatively close to the ice cream cart, and there are no obligations against this ride so they can go as a large group. The walk over takes one minute, and waiting at 4:01 PM is 5 minutes. The ride goes for 3 minutes, so they will finish at 4:09 PM.

Subsequent to the carousel, the group's next ride will be the Pentagon Bumpers. The walk takes 3 minutes, and when they get to the line the time will be 4:12 PM. The waiting time then is 10 minutes, and the ride will last 5 minutes. In total, this activity takes 18 minutes.

After they get off the bumpers, the friend group will go to ride the Dodecagon Rapids. Being one of Freya's priorities, close to their previous destination, and having a relatively small waiting time, this is a great choice. It takes just under one minute to walk there, and once they arrive it will be 4:28 PM. Because it is practically half past 4, the waiting time will be halfway between the waiting time at 4:00 PM and 5:00 PM. This is 7.5 minutes. The ride will go for 10 minutes, elapsing a total of 19 minutes.

After a couple of rides and other activities, the group would most likely need to use the bathroom again. It takes roughly 4 minutes to walk from Dodecagon Rapids to the WC, and they will spend 10 minutes using the restroom. This brings us to 5:00 PM.

Next, they will go to the Hexa-Swings. The walk will be about 3 minutes, and the waiting time at 5:03 PM is 10 minutes. The ride lasts 2 minutes, elapsing a total of 15 minutes.

Moving on to the next ride, the group will proceed to the Square Scramble. This is a medium thrill ride, and suits everyone perfectly fine. Additionally, waiting times are low for this ride. The walk from Hexa-Swings is just under 3 minutes, and upon arrival at 5:18 PM, they must wait approximately 8.75 minutes. The ride lasts 2.5 minutes, so the activity will take roughly 15 minutes in total.

The time is now 5:30 PM. The most suitable ride for this time is the Concave Train. There are no obligations against it, it is nice and easy, and the time setting is completely different from when they went on it in the morning. This means that it would be a completely different experience, with a different view. The walk from the Square Scramble is around 3 minutes, and the waiting time at 5:33 PM is just under 10 minutes. The ride lasts 15 minutes, meaning this activity will end at 5:58 PM.

After the Concave Train, we can conclude that it would be nice to ride the Octagon Flume again, because Freya likes water rides. The walk there will take 4 minutes, and upon arriving at the ride the waiting time will be 10 minutes. The ride lasts 5 minutes. The total time elapsed for this event is 19 minutes.

After the train ride, the group could use another bathroom break. Walking over to the WC will take 2 minutes exactly, and they will spend another 10 minutes there. After finishing using the restroom, the time will be 6:29 PM.

The day is coming to an end, and there are still a few more things we need to fulfill. There is only one more priority to satisfy, and that is Freya's wish to go to the gift shop and get a souvenir. Afterwards, the group must eat dinner. In order to end on a good note, they will watch fireworks in the night sky, ending their day trip to Polygon Paradise Park on a beautiful note.

First things first, they must visit the gift shop. Triangle Restaurant is extremely busy right now, so it would make no sense to waste an hour waiting for dinner. Instead, the group will walk to the gift shop, taking 2 minutes to arrive. Upon arrival, they will spend 30 minutes looking for a suitable gift for themselves, then purchasing it. Half an hour is an extremely reasonable amount of time for a gift shop.

Now it is 7:01 PM and crowds at the Triangle Restaurant have cleared up. The group are bound to head there and eat dinner. The short walk from the gift shop to the Triangle Restaurant takes just one minute, and the group will make their way there at

7:02 PM. The wait time now is just 10 minutes, so they can start eating dinner by 7:12 PM. They spend a good 45 minutes on their dinner, finishing at 7:57 PM. Now, they start walking towards the Trapezoid show for their final activity.

Their walk there takes them 3 minutes, so they arrive just in time for the Firework Show at 8:00 PM. The show will end in an hour at 9:00 PM, which is when their day trip ends and they will head home. This concludes the schedule, and the group's amazing day at Polygon Park!

Time	**Ishmael and Karine**	**Ming and Freya**	**Everyone**
9:00 - 9:07			**Decagon Coaster**
9:07 - 9:25			**Concave Train**
9:25 - 9:33			**Hexa-Swings**
9:33 - 9:39			**Square Scramble**
9:39 - 9:52			**Octagon Flume**
		Arcade (9:52 - 10:34)	
	Heptagon Coaster (9:52 - 10:09)		
	Rhombus Ride (10:09 - 10:22)		
	Rhombus Ride (10:22 - 10:33)		
10:35 - 10:45			**WC**
10:45 - 10:58			**Starburst Carousel**
10:58 - 11:12			**Pentagon Bumpers**
11:12 - 11:30			**Dodecagon Rapids**

11:30 - 11:44			**Pentagon Bumpers**
11:44 - 11:59			**Arcade**
12:00 - 1:00			**Circus Acrobatics**
1:00 - 1:54			**Triangle Restaurant**
1:54 - 3:00			**Magic Show**
3:00 - 3:47			**Arcade**
3:47 - 4:00			**Ice Cream Cart**
4:00 - 4:09			**Starburst Carousel**
4:09 - 4:27			**Pentagon Bumpers**
4:27 - 4:46			**Dodecagon Rapids**
4:46 - 5:00			**WC**
5:00 - 5:15			**Hexa-Swings**
5:15 - 5:30			**Square Scramble**
5:30 - 5:58			**Concave Train**
5:58 - 6:17			**Octagon Flume**
6:17 - 6:29			**WC**
6:29 - 7:00			**Gift Shop**
7:00 - 7:57			**Triangle Restaurant**
7:57 - 9:00			**Firework Show**

5 Analysis

5.1 Strengths

Like any model, our method has strengths and weaknesses. One of our model's strengths is that it is extremely easy to implement and has a simple yet logical structure. Similar to many powerful chess engines, our model does not consider all possible "moves," but rather only the ones deemed the most likely to be optimal. We do this by only considering options that yield an event in which our variables are close to optimal (i.e facilitating short distances and wait times and conforming to the groups preferences).

Another strength of our solution is that it considers everyone's preferences equally in a ranked and organized manner that provides for an equally enjoyable experience for everyone.

Among the specific strengths of our solution are:

- We organized the waiting time for the rides relatively well. This allows the group to conveniently visit all the rides in time.
- We arranged for the group to watch their preferred shows at times when the general waiting times were the highest.
- The group is split up effectively and makes good use of the allotted time
- The group only split up for around 40 minutes, and they spent most of the time together, which was a preference of Freya.
- The group did all the rides twice, except the heptagon and octagon coaster, which proves they made good use of their time and enjoyed all the park had to offer.
- The group was able to go to all the activities they wanted to do.

5.2 Weaknesses

Our model's weakness is that while it produces a fairly good schedule it obviously does not produce the ideal schedule, and one does not know how close it is to the ideal schedule. Also, while as previously mentioned it is very simple to implement, it is slow to apply without the aid of a coded version of the algorithm. Another weakness is that given the limitations of the given data our solution makes extensive use of various assumptions and does not assign numerical values to what is given and what can be deduced.

Among the specific weaknesses of our solution are:

- Ming doesn't split off from the group in the first ride, which is a roller coaster. Ming would get motion sickness, and likely wouldn't enjoy this ride. However, it is not efficient for them to split up at this time, so we have to prioritize the rest of the group over Ming, and she could even choose to sit the ride out.
- Ishmael would probably like to go on the roller coasters several more times.
- The group did not visit the Hot Dog stand, as Karine does not like hotdogs.
- The group

6 Letter to The Group of Friends

Dear Friends,

We are thrilled to be honored with the task of making your experience at Polygon Park the best it can be! Based on your preferences, we have organized a schedule so that everyone can have an amazing experience. Your day starts with the Decagon Coaster, which is a great choice for most of the group. Since this is one of the busiest rides, it's best to go at times where the wait time is minimal. After, it's best for your group to do the 2 rides near the Decagon Coaster to minimize walking time. An hour-long visit to the arcade follows this, which would be a highlight for Ming and "compensation" for the first activity. Next, the schedule has you visiting the Circus Acrobatics show, a must for Karine. After the hour long show, you will be ready for your hour long lunch break. Right after lunch, your group heads to the Magic Show. This is quite convenient as the waiting times of rides rise to a peak at this hour. Right after the show, your group visits the arcade again so everyone can play a few more games while waiting times for rides are still high. By now, quite some time has passed after lunch so it would be a high time to grab a scoop of ice cream at the Ice Cream Cart. After this break, your group attends several rides, including the Starburst Carousel, the Pentagon Bumpers, Hexa-Swings, Rhombus Ride, and the Dodecagon Rapids, one of Freya's priorities. As your trip is coming to a close, it would be a great time for you to visit the souvenir shop and get something to remember your fabulous stay. At around 7 you will be ready for dinner at the Triangle Restaurant, and at 8 you will attend the fireworks show, in awe of both the fireworks and the great time you had at Polygon Park!

We hope you enjoy this schedule,
Team 13296

7 Our Schedule

The above schedule is great for Ming, Ishmael, Karine, and Freya. However, if our group went to Polygon Paradise, our schedule would look different since we have different preferences. Two of our group members don't enjoy water rides, and one can't eat ice-cream. One of our members loves roller coasters, one loves fireworks, and two would love to go to the carousel. All members don't want to go to the arcade, so unlike Ming and the previous group, we will not make a stop at the arcade during the day. On top of that, our group wouldn't want to split up since we would all find it more enjoyable with friends, so we would have to make compromises and might have to sit out on a couple rides. We are assuming all other variables (walking speed, distance, and wait times) would stay constant. Our schedule would consist of starting at Heptagon Coaster and going around the park, skipping the arcade and water rides, to minimize walk time. During lunch time, we would spend an hour at the triangle restaurant, since that is the minimum amount of time we would like to spend in a restaurant to have full enjoyment. We would then have a bathroom break and try the other rides, riding on the carousel a couple of times when the wait time is low. During our trip, Our group would end the day looking at the beautiful fireworks show.

Just like we applied this model to create a schedule for our group, this model can be applied to create a schedule with different parameters. Using the same method of considering the next optimal event based on preferences, distance, and wait times, we can develop a schedule regardless of the specific parameters.

8 Sources and Works Cited

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